



# SENDER TECHNICAL INTEGRATION GUIDE

API Specifications • Security • Workflow • JSON Schema

## DOCUMENT PARTICULARS

|                   |                                                                             |
|-------------------|-----------------------------------------------------------------------------|
| <b>Receiver:</b>  | Alshumookh Global Banking Financing And Credit                              |
| <b>Platform:</b>  | <a href="https://api.alshumookh-pay.com">https://api.alshumookh-pay.com</a> |
| <b>Issued By:</b> | ALSHUMOOKH GROUP — Technology & Compliance                                  |
| <b>Purpose:</b>   | API-to-API Technical Integration Reference                                  |
| <b>Date:</b>      | May 2026   STRICTLY CONFIDENTIAL                                            |

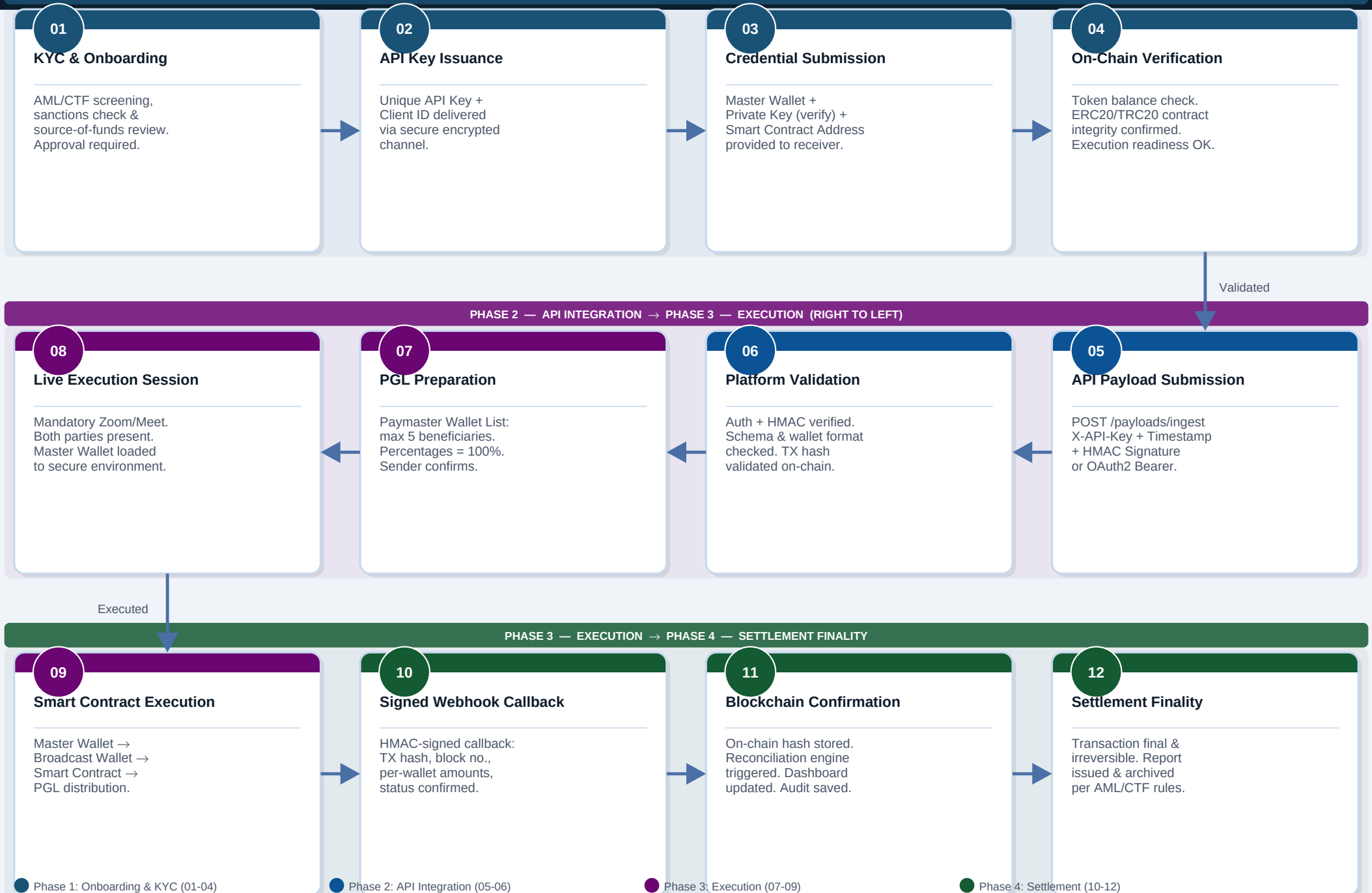
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# Complete Sender-to-Receiver Settlement Workflow

<https://api.alshumookh-pay.com>

12-step end-to-end process: Onboarding → API Integration → Execution → Settlement Finality



DONE  
Final

01

## RECEIVER PROFILE

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| Entity Name     | Alshumookh Global Banking Financing And Credit                       |
| Role            | Receiver / Execution Party — Digital Asset Transactions              |
| Jurisdiction    | Dubai, United Arab Emirates                                          |
| Office Address  | API World Tower, Office No. 2103/2104, Shaikh Zayed Road, Dubai, UAE |
| License No.     | 887065                                                               |
| Authorised By   | Ahmed Mohamed — Chief Executive Officer                              |
| Contact Email   | ceo@alshumookhgroup.ae                                               |
| Platform URL    | https://api.alshumookh-pay.com                                       |
| Ingest Endpoint | POST https://api.alshumookh-pay.com/api/v1/payloads/ingest           |
| OAuth2 Token    | POST https://api.alshumookh-pay.com/api/v1/oauth/token               |
| Receiver Wallet | 0xBD682cfD8382a90adfDd6745780D3D7959c4d939                           |

02

## SECURITY ARCHITECTURE — ALL ACTIVE CONTROLS

|                                                                                                                                    |                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>HMAC-SHA256</b><br>Payload Signing       | <br><b>OAuth 2.0</b><br>Client Credentials      |
| <br><b>JWT / JWS</b><br>Crypto Verification     | <br><b>mTLS Ready</b><br>Cert Fingerprint       |
| <br><b>Signed Webhooks</b><br>S2S Notifications | <br><b>Reconciliation</b><br>Full Audit Engine  |
| <br><b>WAF Shield</b><br>Intrusion Protection   | <br><b>TLS 1.3 / HSTS</b><br>Transport Security |

## Security Controls — Technical Specification

| Control          | Header / Protocol         | Technical Detail                                                                       |
|------------------|---------------------------|----------------------------------------------------------------------------------------|
| HMAC-SHA256      | X-Signature               | HMAC-SHA256(X-Timestamp + '.' + raw_body). Enforced per-client flag.                   |
| OAuth 2.0        | Authorization: Bearer     | client_credentials grant at POST /api/v1/oauth/token.                                  |
| JWT / JWS        | X-JWS-Signature           | Detached JWS with RS256/HS256 via PyJWT. Payload hash in claims.                       |
| mTLS Cert        | X-Client-Cert-Fingerprint | SHA-256 cert fingerprint forwarded by proxy, verified at app layer.                    |
| Signed Webhooks  | X-ALSHUMOOKH-Signature    | All outbound callbacks HMAC-signed. Alchemy / Coinbase / MoonPay inbound verification. |
| Reconciliation   | POST /admin/reconcile     | Full audit trail: tx hash, amounts, timestamps, status per payload.                    |
| WAF / Rate Limit | Middleware                | IP auto-ban, user-agent classification, path-probe detection.                          |
| Replay Guard     | X-Timestamp               | Timestamp validated $\pm 5$ min window. Idempotency-Key enforced.                      |
| Transport        | TLS 1.2 / 1.3             | HSTS max-age=31536000. CSP, X-Frame-Options, X-Content-Type-Options.                   |
| IP Whitelist     | Middleware                | Per-client allowed IP ranges enforced at gateway layer.                                |

## Authentication Modes — Configurable Per Client

| Level | Mode           | Required Headers                                  | Grade         |
|-------|----------------|---------------------------------------------------|---------------|
| 1     | API Key Only   | X-API-Key + X-Client-ID                           | Standard      |
| 2     | API Key + HMAC | X-API-Key + X-Signature + X-Timestamp             | Enhanced      |
| 3     | OAuth2 Bearer  | Authorization: Bearer <token>                     | Professional  |
| 4     | OAuth2 + HMAC  | Authorization: Bearer + X-Signature + X-Timestamp | Institutional |
| 5     | mTLS + Key     | X-Client-Cert-Fingerprint + X-API-Key             | Maximum       |
| 6     | OAuth2 + JWS   | Authorization: Bearer + X-JWS-Signature           | Full Crypto   |

All authentication modes are provisioned per client at the API-key level. The security tier is assigned based on transaction volume and institutional requirements.

### 03

## TRANSACTION WORKFLOW — END TO END

Complete process from sender onboarding through blockchain settlement finality. All 12 steps are mandatory and executed in sequence.

|    |                          |                              |                                                                                                                                                                                                                |
|----|--------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <input type="checkbox"/> | <b>KYC &amp; ONBOARDING</b>  | Sender submits KYC documentation. ALSHUMOOKH Compliance conducts AML/CTF screening, sanctions check, and source-of-funds review. Internal approval granted before any operation.                               |
| 02 | <input type="checkbox"/> | <b>API KEY ISSUANCE</b>      | Upon KYC approval, Sender receives API Key + Client ID via secure encrypted channel. System integration is configured pointing to: <a href="https://api.alshumookh-pay.com">https://api.alshumookh-pay.com</a> |
| 03 | <input type="checkbox"/> | <b>CREDENTIAL SUBMISSION</b> | Sender provides: Master Wallet Address + Private Key (technical verification only) + Smart Contract Address. This authorises Receiver to conduct controlled on-chain verification.                             |

|    |                          |                          |                                                                                                                                                                                                                                                   |
|----|--------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04 | <input type="checkbox"/> | ON-CHAIN VERIFICATION    | ALSHUMOOKH verifies: token balance, USDT ERC20/TRC20 smart contract integrity, and execution readiness. No execution proceeds until all on-chain checks pass.                                                                                     |
| 05 | <input type="checkbox"/> | API PAYLOAD SUBMISSION   | Sender POSTs signed JSON payload to: <a href="https://api.alshumookh-pay.com/api/v1/payloads/ingest">https://api.alshumookh-pay.com/api/v1/payloads/ingest</a> Must include: X-API-Key + X-Timestamp + X-Signature (HMAC) or OAuth2 Bearer token. |
| 06 | <input type="checkbox"/> | PLATFORM VALIDATION      | System validates: API Key/Bearer token, HMAC signature, sender_reference uniqueness, wallet format, transaction_hash on-chain, timestamp window (±5 min), and JSON schema.                                                                        |
| 07 | <input type="checkbox"/> | PGL PREPARATION          | Receiver prepares Payout/Paymaster Wallet List (PGL) — max 5 beneficiary wallets with percentage distribution totalling exactly 100%. Sender confirms in writing.                                                                                 |
| 08 | <input type="checkbox"/> | LIVE EXECUTION SESSION   | Mandatory live session via Zoom or Google Meet. Final authorisation checkpoint. Both parties present. Master Wallet imported into controlled secure environment.                                                                                  |
| 09 | <input type="checkbox"/> | SMART CONTRACT EXECUTION | Tokens transferred: Master Wallet → Bank Broadcast Wallet → Smart Contract. Distribution executed automatically per approved PGL percentages.                                                                                                     |
| 10 | <input type="checkbox"/> | SIGNED WEBHOOK CALLBACK  | HMAC-signed webhook dispatched to Sender's endpoint confirming: execution status, transaction hash, block number, and per-wallet distribution amounts.                                                                                            |
| 11 | <input type="checkbox"/> | BLOCKCHAIN CONFIRMATION  | On-chain confirmation hash recorded. Transaction receipts retained for compliance audit. Dashboard updated. Reconciliation engine triggered automatically.                                                                                        |
| 12 | <input type="checkbox"/> | SETTLEMENT FINALITY      | Transaction deemed final and irreversible upon on-chain confirmation. Formal settlement report issued. Records archived per AML/CTF requirements.                                                                                                 |

**SETTLEMENT COMPLETE — Final and irreversible upon on-chain confirmation. HMAC-signed settlement report dispatched to Sender's registered webhook endpoint.**

04

## COMPLETE ACCEPTED JSON PAYLOAD — FULL SCHEMA

Every request to the ingest endpoint must include a JSON body matching the schema below. Fields marked ★ are mandatory. Amounts are decimal (float). All strings are UTF-8.

## 4.1 Field Reference

| Field               | Type   | ★ | Description / Accepted Values                                    |
|---------------------|--------|---|------------------------------------------------------------------|
| sender_reference    | string | ★ | Unique TX ID from Sender. Max 128 chars. Rejected if duplicate.  |
| client_id           | string | ★ | Client ID issued by ALSHUMOOKH at onboarding.                    |
| amount              | number | ★ | Decimal amount > 0. Example: 1000000.00                          |
| currency            | string | ★ | Fixed value: "USDT"                                              |
| network             | string | ★ | Must be: "ERC20" or "TRC20"                                      |
| sender_wallet       | string | ★ | Master wallet. ERC20: 0x... (42 chars). TRC20: T... (34 chars).  |
| receiver_wallet     | string | ★ | ALSHUMOOKH wallet: 0xBD682cfD8382a90adfDd6745780D3D7959c4d939    |
| smart_contract      | string | ★ | Smart contract address used for token execution.                 |
| transaction_hash    | string | ★ | On-chain TX hash — must be verifiable on the specified network.  |
| beneficiary_wallets | array  | ★ | Array of objects. Min 1, max 5. All percentages must sum to 100. |
| ■ wallet            | string | ★ | Beneficiary wallet address (inside beneficiary_wallets).         |
| ■ percentage        | number | ★ | Distribution %. Integer or decimal. All items must sum to 100.   |
| timestamp           | string | ★ | ISO-8601 UTC. e.g. 2026-05-11T10:00:00Z. Within ±5 minutes.      |
| notes               | string | — | Optional. Free text. Max 500 chars.                              |
| webhook_url         | string | — | Optional HTTPS callback URL for signed status notifications.     |
| idempotency_key     | string | — | Recommended. Prevents duplicate processing on retry.             |

## 4.2 Complete JSON Example — All Fields

```
{ "sender_reference": "TXN-2026-001", "client_id": "YOUR_CLIENT_ID", "amount": 1000000.00, "currency": "USDT", "network": "ERC20", "sender_wallet": "0xYOUR_MASTER_WALLET_ADDRESS", "receiver_wallet": "0xBD682cfD8382a90adfDd6745780D3D7959c4d939", "smart_contract": "0xSMART_CONTRACT_ADDRESS", "transaction_hash": "0xBLOCKCHAIN_TX_HASH", "beneficiary_wallets": [ { "wallet": "0xWALLET_ADDRESS_1", "percentage": 40 }, { "wallet": "0xWALLET_ADDRESS_2", "percentage": 35 }, { "wallet": "0xWALLET_ADDRESS_3", "percentage": 25 } ], "timestamp": "2026-05-11T10:00:00Z", "notes": "Optional transaction notes", "webhook_url": "https://your-system.com/webhook/callback", "idempotency_key": "IDEM-2026-001" }
```

## 4.3 Minimal JSON Example — Required Fields Only

```
{ "sender_reference": "TXN-2026-001", "client_id": "YOUR_CLIENT_ID", "amount": 1000000.00, "currency": "USDT", "network": "ERC20", "sender_wallet": "0xYOUR_MASTER_WALLET_ADDRESS", "receiver_wallet": "0xBD682cfD8382a90adfDd6745780D3D7959c4d939", "smart_contract": "0xSMART_CONTRACT_ADDRESS", "transaction_hash": "0xBLOCKCHAIN_TX_HASH", "beneficiary_wallets": [ { "wallet": "0xWALLET_ADDRESS_1", "percentage": 60 }, { "wallet": "0xWALLET_ADDRESS_2", "percentage": 40 } ], "timestamp": "2026-05-11T10:00:00Z" }
```

### 05

## HTTP REQUEST EXAMPLES

### 5.1 Standard Request — API Key Authentication

```
POST https://api.alshumookh-pay.com/api/v1/payloads/ingest Content-Type: application/json X-API-Key: YOUR_API_KEY X-Client-ID: YOUR_CLIENT_ID X-Timestamp: 2026-05-11T10:00:00Z X-Idempotency-Key: IDEM-2026-001
Body: { "sender_reference": "TXN-2026-001", "amount": 1000000.00, ... }
```

## 5.2 Institutional Request — OAuth2 Bearer + HMAC-SHA256

```
# STEP 1 – Obtain Bearer Token POST https://api.alshumookh-pay.com/api/v1/oauth/token Content-Type:
application/x-www-form-urlencoded Body: grant_type=client_credentials&client_id=ID&client_secret=SECRET
Response: {"access_token": "eyJ...", "token_type": "bearer", "expires_in": 3600} # STEP 2 – Compute HMAC
Signature SIGNATURE = HMAC-SHA256( X-Timestamp + "." + raw_json_body, hmac_secret ) # STEP 3 – Submit
Payload POST https://api.alshumookh-pay.com/api/v1/payloads/ingest Content-Type: application/json
Authorization: Bearer eyJ... X-Timestamp: 2026-05-11T10:00:00Z X-Signature: X-Client-ID: YOUR_CLIENT_ID
Body: { ... payload body ... }
```

## 5.3 API Response Codes

| HTTP | Status   | Error Code          | Meaning                                 | Action Required               |
|------|----------|---------------------|-----------------------------------------|-------------------------------|
| 200  | accepted | –                   | Payload received — processing initiated | Await webhook callback        |
| 422  | error    | validation_error    | Missing or invalid field in payload     | Fix field and resubmit        |
| 401  | error    | invalid_api_key     | API Key missing or invalid              | Check X-API-Key header        |
| 401  | error    | hmac_required       | X-Signature header missing              | Add HMAC-SHA256 signature     |
| 401  | error    | invalid_hmac        | HMAC signature does not match           | Verify signature computation  |
| 401  | error    | oauth_required      | OAuth2 Bearer token required            | Obtain token via /oauth/token |
| 409  | error    | duplicate_reference | sender_reference already processed      | Use a new unique reference    |
| 429  | error    | rate_limited        | Too many requests                       | Retry after 60 seconds        |
| 500  | error    | server_error        | Internal server error                   | Contact support immediately   |

## 5.4 Outbound Signed Webhook — Callback to Sender

```
POST https://your-system.com/webhook/callback Content-Type: application/json X-ALSHUMOOKH-Signature:
X-Timestamp: 2026-05-11T10:05:00Z { "event": "payload.settled", "payload_id": "PLY-XXXX",
"sender_reference": "TXN-2026-001", "status": "settled", "amount": 1000000.00, "currency": "USDT",
"network": "ERC20", "transaction_hash": "0xCONFIRMED_TX_HASH", "block_number": 19500000, "settled_at":
"2026-05-11T10:04:55Z", "distribution": [
{"wallet": "0xWALLET_A", "amount": 400000, "percentage": 40, "tx_hash": "0x..."},
{"wallet": "0xWALLET_B", "amount": 350000, "percentage": 35, "tx_hash": "0x..."},
{"wallet": "0xWALLET_C", "amount": 250000, "percentage": 25, "tx_hash": "0x..."} ] }
```

## 5.5 Payload Validation Rules

| Field / Rule        | Condition                                           | Error Code if Violated |
|---------------------|-----------------------------------------------------|------------------------|
| sender_reference    | Unique per client — previously seen values rejected | duplicate_reference    |
| amount              | Positive decimal number greater than 0              | invalid_amount         |
| currency            | Must be exactly "USDT"                              | invalid_currency       |
| network             | Must be "ERC20" or "TRC20"                          | invalid_network        |
| beneficiary_wallets | 1–5 items. All percentages must sum to exactly 100  | invalid_pgl            |
| transaction_hash    | Must resolve on-chain for the specified network     | unverifiable_hash      |
| timestamp           | Must be within ±5 minutes of server UTC time        | timestamp_expired      |
| receiver_wallet     | Must exactly match ALSHUMOOKH's registered wallet   | wallet_mismatch        |
| X-Signature         | Required when client flag hmac_required = true      | hmac_required          |



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